

1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives

Generative AI is increasingly used to design novel drug candidates, but current workflows rely on static molecular descriptors and computational proxies for potency and safety, not on how a molecule actually behaves in vivo -- absorption, distribution, metabolism, and the resulting concentration-time profile. Physiologically based pharmacokinetic (PBPK) modeling can predict this behavior mechanistically, but is currently used almost exclusively as a downstream evaluation step, after candidates have already been generated. This project's central, verifiable objective is to make PBPK an explicit optimization signal inside generative molecular design, rather than a downstream check -- assessed concretely through three measurable objectives (Section 1.2) and validated across three target families (Section 3.1).

Beyond the state of the art: PBPK-informed pharmacology assessment is established in medicinal chemistry, and recent work (Van Rompaey et al., 2024) has used allometrically-scaled animal PK to triage designs pre-synthesis, but for potency only, and pre-synthesis rather than inside the generative loop. To our knowledge, no existing work incorporates mechanistic, time-resolved PBPK simulation directly into generative optimization for both potency and safety jointly. The applicant's own preliminary audit (Section 1.2) provides direct, quantitative evidence that the current state of the art -- static filters and composite scoring -- systematically fails to surface pharmacokinetically favorable candidates, which is the specific gap this project is designed to close.

1.2 Soundness of the proposed methodology

Preliminary results (DAAD-funded work at ICSI Berkeley): using a PBPK simulation layer (OSPSuite/PK-Sim, ADMET-AI-derived ADME parameters, Chemprop D-MPNN potency/toxicity predictors), the applicant audited three structurally distinct PARP1-targeted generative pipelines against their own conventional, static potency and safety metrics. Despite passing every conventional static filter, a subset of shortlisted candidates had no pharmacokinetically achievable dose at all; among those that did, roughly 70% could not be confidently classified as safe once prediction and simulation uncertainty was propagated, and about 85% were each simultaneously outperformed on potency and on both safety margins by an independently-dosed alternative that conventional composite scoring had not selected. The remainder of the DAAD-funded stay will extend this audit itself, rather than begin the closed-loop optimization below; the trainable surrogate and closed-loop integration are new work that begins only with MSCA funding.

Objective 1 -- a PBPK-aware molecular optimization framework: connect generative molecular models to mechanistic PBPK simulation via a trainable surrogate. Since full PBPK simulation is too slow to run inside a generative sampling loop, a graph neural network (consistent with the Chemprop D-MPNN architecture already validated in the preliminary work) will be trained to predict PBPK-simulation-derived quantities (MED, C_{max}, C_{ss,min}) in place of a full PK-Sim run at each generative step. Training data bootstraps from the approximately 1,979 pool-scale PBPK simulations already generated during the preliminary audit, supplemented by active-learning-selected simulations near decision boundaries. Surrogate accuracy will be validated by held-out correlation/RMSE against full PK-Sim simulation, and by comparing surrogate-guided optimization outcomes against a full-simulation-in-the-loop run on a validation subset.

Objective 2 -- time-resolved pharmacokinetic optimization: represent the full concentration-time profile (C_{max}, C_{ss,min}, time-above-threshold), not a single point estimate, as the optimization target. The preliminary DAAD-funded work relied on a single virtual individual and did not model inter-individual pharmacokinetic variability; this objective is developed jointly with the return host's population PK/PD and nonlinear mixed-effects expertise (Section 1.3) to incorporate that variability directly into the exposure objective -- a specific methodological limitation of the outgoing-phase work, not a generic addition.

Objective 3 -- generalizability and biological relevance: evaluate the framework across three target families -- PARP1 from the preliminary work, JAK2 (added during the outgoing phase; different dose-limiting toxicity and clearance mechanism, reusing existing ChEMBL/DOCKSTRING data), and a third, structurally distinct target added during the return phase once the framework is already validated. A single additional target would leave the generalizability claim resting on an n of two; a third strengthens it without adding risk to the tighter 18-month outgoing-phase timeline.

Interdisciplinary integration: the project structurally requires two disciplines that are not usually combined -- generative machine learning (outgoing phase) and mechanistic pharmacometrics/nonlinear mixed-effects modeling (return phase) -- integrated through a shared, quantitative interface (the PBPK surrogate and its exposure objective), not merely two separate work packages run in sequence.

Gender dimension: this is a computational methods project without human or animal subjects; sex/gender is not a biologically relevant variable in the modeling itself, and no gender dimension is considered applicable to the research content.

Open science practices: code and, where licensing permits, simulation workflows will be released openly, continuing the practice already followed in the preliminary DAAD-funded work; a data management plan will be developed within the first 6 months covering the PBPK simulation outputs and trained surrogate models as the project's main research data.

1.3 Quality of supervision, training, and three-way transfer of knowledge

As a Global Fellowship, this project involves a three-way transfer of knowledge between the researcher, the return host (University of Pavia), and the associated partner for the outgoing phase (ICSI Berkeley). The outgoing phase, under the supervision of Dr. Benjamin Erichson, builds advanced computational skills: generative modeling architectures, reinforcement learning and surrogate-model/active-learning methods, and large-scale simulation pipelines. The return phase, under the supervision of Prof. Paolo Magni, closes a specific gap rather than adding general pharmacometrics exposure: population PK/PD and nonlinear mixed-effects modeling, which the preliminary DAAD-funded work does not use. This combination establishes an independent research profile at the intersection of generative AI and quantitative pharmacology, extending directly from the applicant's PhD work on PBPK modeling of large molecules (RWTH Aachen, with Boehringer Ingelheim) and current DAAD-funded work on generative drug design (ICSI Berkeley).

Dr. N. Benjamin Erichson (Senior Research Scientist and Research Group Leader, Robust Deep Learning Group, ICSI; also affiliated with Lawrence Berkeley National Laboratory) holds a PhD in Statistics (University of St Andrews) and previously held positions as a tenure-track assistant professor (University of Pittsburgh) and postdoctoral researcher (UC Berkeley Statistics; RISELab, UC Berkeley EECS; University of Washington Applied Mathematics). His research on the intersection of deep learning, dynamical systems and robustness is directly relevant to the surrogate-model methodology at the core of Objective 1.

Prof. Paolo Magni (Full Professor of Bioengineering, University of Pavia; PhD in Bioengineering, 1998) is Director of the Bioinformatics, Mathematical Modelling and Synthetic Biology (BMS) Laboratory and has taught mathematical modeling of biological systems, physiology, statistics, bioinformatics and pharmacometrics since 2005. He is author of more than 400 publications (160 in international journals), co-founder of two spin-off companies, and co-founder of the DDMoRe Foundation, which originated from the FP7-IMI DDMoRe project that he co-led as UNIPV PI and work-package leader. Since 2004 he has supervised 42 PhD students (several awarded the Italian national prize for best Bioengineering PhD thesis) and 16 postdoctoral researchers, including 4 PhD students hosted from other universities and countries, with international collaborations spanning Pamplona, Paris, Berlin, Athens, London, Leiden, Valencia and Porto Alegre. He has consulted for major pharmaceutical companies (including GSK, Janssen, Chiesi, Merck Serono, Menarini, Italfarmaco, Nerviano Medical Science, Servier, Angelini and

Pharmacia) on PK/PD population modeling, PBPK modeling, and Bayesian/MCMC estimation methods. This would be his first MSCA-funded fellow, extending an already substantial doctoral and postdoctoral supervision record to this specific scheme.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

The applicant holds a PhD (Dr. rer. nat., RWTH Aachen University, awarded 15/12/2025, overall grade "sehr gut"), supervised by Univ.-Prof. Dr.-Ing. Lars Mathias Blank and Univ.-Prof. Dr. rer. nat. Lars Kupfer, conducted in collaboration with Boehringer Ingelheim on PBPK modeling of large molecules for retinal delivery, providing direct mechanistic PBPK modeling experience prior to this project. The current DAAD-funded postdoctoral stay at ICSI Berkeley has built the generative-AI and computational drug-discovery skills demonstrated in the preliminary audit (Section 1.2): PBPK simulation pipelines (OSPSuite/PK-Sim), ADMET-AI, and Chemprop-based potency/toxicity modeling. This combination of mechanistic PBPK experience (PhD) and generative-AI/computational drug-discovery experience (current postdoc) is the specific, non-redundant background the proposed project builds on.

[TO CONFIRM: full CV-level detail (exact dates, additional skills/competences) is in Part B2 Section 4 -- keep this subsection focused on relevance to the project, not a CV duplicate]

2. Impact

2.1 Credibility of measures to enhance career perspectives and employability

The outgoing phase develops advanced, transferable computational skills (generative modeling, surrogate/active-learning methods) increasingly sought in both academic and industry drug-discovery roles; the return phase develops mechanistic pharmacometric skills (population PK/PD, nonlinear mixed-effects modeling) that are comparatively scarce among AI-trained researchers. Together, these position the applicant for an independent research-leadership role at the intersection of generative AI and quantitative pharmacology, in academia or in pharmaceutical R&D, where this combination of skills is currently rare.

[TO CONFIRM: concrete transferable-skills training activities (e.g., specific courses, schools, grant-writing or supervision training offered by ICSI/Pavia) to be added once confirmed with both supervisors]

2.2 Dissemination, exploitation and communication plan

Results will be submitted to venues spanning both communities: machine-learning-for-drug-discovery venues (e.g. NeurIPS AI4DD, ICML MLDD workshops) and pharmacometrics/quantitative pharmacology venues (e.g. PAGE, ACoP, journals such as CPT: Pharmacometrics & Systems Pharmacology or the Journal of Cheminformatics), reaching both the scientific community and, through ICSI's and Pavia's own communication channels, a broader audience. Code and, where licensing permits, simulation workflows will be released openly to support reproducibility and reuse by other groups, continuing the practice already followed in the preliminary DAAD-funded work.

Intellectual property: this is a methods/software-focused project; no patentable output is currently anticipated, and open-source release is the default exploitation route. Should a specific, protectable output emerge (e.g., a deployable surrogate model of independent commercial interest), IP management would follow the beneficiary institution's standard technology-transfer procedures.

[TO CONFIRM: specific target journals/conferences and communication channels to finalize with both supervisors; confirm IP position with Pavia's technology transfer office if relevant]

2.3 Magnitude and importance of the project's contribution to expected impacts

Scientific impact: a computational framework integrating generative AI with mechanistic PBPK modeling, plus a systematic, quantified characterization of where conventional potency/safety objectives fail to predict PBPK-confirmed outcomes -- directly evidenced by the preliminary audit, where 85% of an already-filtered, composite-score-selected shortlist was dominated by an unselected alternative on every PBPK dimension simultaneously.

Economic/technological impact: fewer computationally promising but pharmacokinetically unsuitable candidates advancing to later, substantially more expensive development stages. Even a modest reduction in late-stage PK-driven attrition -- the specific failure mode this project targets -- translates into material cost and time savings in pharmaceutical R&D pipelines that already rely on generative design. The outgoing phase's location in the San Francisco Bay Area -- the world's largest biotech cluster (200+ companies, USD 142.7B revenue in 2023) -- further supports future translational engagement and dissemination of the resulting framework to industry.

Societal impact: faster, more reliable early-stage triage of AI-generated drug candidates supports the broader goal of accelerating safe therapeutics to patients, by reducing the share of computational effort spent on candidates that were never going to be viable once real pharmacokinetics are taken into account.

[TO CONFIRM: quantified estimates beyond the preliminary audit's own numbers (e.g., an estimated cost-per-candidate saving) would strengthen this section but require input/benchmarks from Pavia or an industry contact -- do not invent a number without a source]

3. Quality and Efficiency of the Implementation

3.1 Work plan, risk assessment, and effort per work package

Feasibility: WP1 does not start from zero. The surrogate model bootstraps from the approximately 1,979 pool-scale PBPK simulations already generated during the preliminary audit, and the remainder of the DAAD-funded stay is scoped to extend that audit rather than pre-empt WP1/WP2's optimization work (Section 1.2). The 18-month outgoing phase is therefore spent on integration and closed-loop optimization, not on building simulation infrastructure or a training dataset from scratch.

WP1 (ICSI, M1-M9): build the computational interface between generative models and PBPK simulation outputs; train the PK surrogate model. Milestone M6: surrogate validated against held-out PK-Sim simulations.

WP2 (ICSI, M6-M18): integrate the time-resolved exposure objective into generative optimization; benchmark against PARP1 and the JAK2 target (Objective 3). Milestone M18: closed-loop framework validated on two target families; secondment/transition to Pavia.

WP3 (Pavia, M19-M24): incorporate population PK/PD and inter-individual variability into the optimization objective (Objective 2), under Prof. Magni's supervision; model validation against literature/experimental PK data where available. Milestone M24: variability-aware objective validated.

WP4 (Pavia, M25-M30): apply the validated framework to a third, structurally distinct target (Objective 3); dissemination and final synthesis/reporting. Milestone M30: three-target generalizability result finalized; final deliverables submitted.

Gantt chart (months elapsed from project start)

WP	Host	M1-6	M7-12	M13-18	M19-24	M25-30
WP1	ICSI					
WP2	ICSI					
WP3	Pavia					
WP4	Pavia					

Milestones: M6 (surrogate validated), M18 (two-target closed-loop framework complete), M24 (variability-aware objective validated), M30 (three-target generalizability result, final reporting).

Risk assessment

- Surrogate PK model insufficiently accurate for optimization feedback (medium likelihood, medium severity) -> mitigate via active learning focused on decision-relevant regions, and fallback to slower

offline PBPK-in-the-loop re-ranking rather than a fully differentiable surrogate.

- Computational cost of large-scale PBPK simulation (low likelihood, medium severity) -> mitigate by reusing the preliminary work's existing simulation database and prioritizing simulation budget via uncertainty-guided sampling.
- Cross-institutional/mobility timeline risk (low likelihood, high severity) -> mitigate via early confirmation of both hosts' administrative deadlines and a fixed, pre-agreed outgoing/return split.

3.2 Quality and capacity of the host institutions, including hosting arrangements

Outgoing host (ICSI Berkeley, associated partner): provides the generative-modeling and computational infrastructure environment in which the preliminary work was carried out, under the supervision of Dr. Benjamin Erichson; the researcher is already integrated in this environment via the current DAAD-funded stay. Full institutional capacity details are provided in Part B2, Section 5.2.

Return host (University of Pavia, beneficiary): provides population PK/PD and nonlinear mixed-effects modeling expertise, under the supervision of Prof. Paolo Magni, that is specifically absent from the outgoing-phase methodology, within the Bioinformatics, Mathematical Modelling and Synthetic Biology (BMS) Laboratory. The researcher will have a dedicated desk in the BMS lab with daily interaction with the lab team, weekly progress meetings, a personal workstation with the required modeling software, and access to the wider engineering faculty's facilities; computational infrastructure includes a departmental cluster and access to the CINECA Leonardo EuroHPC supercomputer. Full institutional capacity details are provided in Part B2, Section 5.2.